

RUHAAN KAKAR

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EDUCATION

Degree	Institute	CGPA / Percentage	Year
B.Tech. (Hons.)	Indian Institute of Technology, Kharagpur	8.48 / 10	2026
Class XII, CBSE	CBSE Board	93.2%	2024
Class X, CBSE	CBSE Board	96.4%	2022

INTERNSHIPS

Bloomington University | Research Intern | *Prof. Da Yan* Jun '26 – Present

- Built real-time aerial perception pipeline with fine-tuned **YOLOv8** and **SORT** tracker for drone-based search-and-rescue surveillance.
- Curated **5K-frame/32K-box** drone dataset; fine-tuned YOLOv8m to **73.4% mAP@0.5**, beating COCO zero-shot baseline by **15%**.
- Deployed CPU-based SORT tracker with Kalman filters achieving **84.7% MOTA** at **28 FPS** without DeepSORT's ReID overhead.
- Exported PyTorch model to **ONNX** with dynamic batching, reducing end-to-end inference latency to **35ms** on standard laptop hardware.

IIT Kharagpur | Research Intern | *Prof. Hari V Warrior* Jul '25 – Sep '25

- Engineered a **(64–32) Stacked LSTM** from 16 sensor inputs for 3D thrust vectoring and emergency braking on AUV trajectories.
- Outperformed XGBoost benchmark on test runs with **22% lower thrust RMSE** and **12% higher braking F1-score** overall.
- Extracted 5 critical physics features using **SHAP importance** from 16 channels, boosting close range collision avoidance by **15%**.
- Cut false negatives by **40%** using Huber + weighted BCE loss for <5% brake events, sustaining **sub-30ms** edge CPU inference.

Bhavya Haulage Services FZCO | Full Stack Developer Intern | *Dubai, UAE* Mar '25 – May '25

- Architected a full-stack enterprise platform unifying **fleet management** and integrated investment portfolio solutions.
- Engineered dynamic real-time **analytics dashboards** and redesigned the client-facing UI for operational visibility.
- Built a robust **Node.js/Express** backend with secure REST APIs, **JWT-based MFA**, and session management systems.

PROJECTS

Fiona – Campus Mental Health Social Platform | Perplexity Hackathon Mar '26 – May '26

- 🔗 <https://fiona-campus.onrender.com/>
- Built a real-time **Next.js/Socket.io** platform with Redis pub/sub, handling **5K users** via $\mathcal{O}(1)$ hash routing for **sub-10ms** delivery.
 - Structured a 2-phase scoring worker computing **VADER sentiment** inline, offloading heavy metrics to a **BullMQ-backed** queue.
 - Prevented alert fatigue using a **7-day OLS regression** slope over rolling windows, with a weighted moving average for anomaly scores.
 - Implemented client-side **Hinglish toxicity filtering** via a 5-token sliding regex trie, cutting moderation latency to **sub-10ms**.

GraphMind – RAG Pipeline for MetaKGP Wiki | DevSoc Hackathon Jan '26 – Feb '26

- Engineered **5.5K-page** RAG vector search engine with **ChromaDB & PostgreSQL**, delivering top-5 search results under **500ms**.
- Developed a **3-agent Mixture of Experts (MoE)** pipeline featuring consensus verification for reliable, hallucination-free responses.
- Accelerated vector retrieval via targeted **HNSW index** tuning & embedding caching, slashing query latency by **35%** under heavy load.
- Architected a scalable **FastAPI** backend with **Groq LLM** inference and serverless **Modal** embeddings for real-time campus-wide Q&A.

TypeBolt – Full-Stack Typing Speed Platform | Self Project May '26 – Jul '26

- 🔗 <https://typebolt.vercel.app/>
- Built a full-stack typing platform using **React, Node.js, and Docker**, enabling consistent, reproducible development environments.
 - Reduced dashboard query latency by **90%** (150ms to under 15ms) using **Redis caching** with TTL-based cache invalidation.
 - Accelerated paginated leaderboard retrieval by **85%** using **compound MongoDB indexes**, eliminating costly in-memory sorting.

AWARDS & ACHIEVEMENTS

- **Gold Medal, Open IIT 2026:** Secured 1st place in Netflix Data Analytics Challenge by TSG, IIT KGP among 22 halls.
- **Competitive Programming:** Solved **500+** DSA problems across LeetCode, CodeChef, AtCoder; cleared internal SIH round.

SKILLS & EXPERTISE

Languages: C, C++, Python, JavaScript, TypeScript, SQL, HTML, CSS, Bash, $\text{\LaTeX}$, Markdown
Frameworks & Libraries: React, Next.js, Node.js, Express, PyTorch, TensorFlow, Scikit-Learn, XGBoost, Pandas, NumPy, SciPy
Tools & Databases: Docker, Git, Redis, MongoDB, PostgreSQL, ChromaDB, ONNX, Vercel, Modal, Jupyter, MATLAB, ANSYS

EXTRA CURRICULAR ACTIVITIES & LEADERSHIP

- **Leadership:** Development Head at Developers Society; Researcher at TeamKART; Secretary at Communique & NAROE.
- **Cultural & Arts:** **Gold** in Stage Play at Thomso (IIT Roorkee); **Silver** in Street Play at Spring Fest; Stage Actor at Encore.
- **Sports & Service:** State-level (Punjab) Badminton Singles; Tertiya Sopan in Bharat Scouts; Organised 3rd GMUN (250+ delegates).